

src/app/docWorkspaceMetrics.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T04:41:43.744Z

Overview

`src/app/docWorkspaceMetrics.ts` is a new module that aggregates file-change statistics and formats them for LLM prompts, Markdown reports, and Mermaid visualizations. The file adds imports, interfaces, helper functions, and several public utilities.

Key changes

- **Imports** – `import type { ChangeKind, FileChange } from "../types";` (R1).
- **Interfaces** – `ExtensionCount` (R3-R6) and `DocWorkspaceMetrics` (R8-R17) describe metric shapes.
- **Helpers** – `fileExtension` (R21-R29) and `firstSegment` (R31-R35) extract extensions and top-level path segments.
- **Metric builder** – `buildDocWorkspaceMetrics` (R37-R85) tallies counts by kind, extension, depth, and root, caps sample paths at 120, and returns a `DocWorkspaceMetrics` object.
- **Prompt formatter** – `metricsToPromptContext` (R93-R143) produces a structured prompt string, optionally including gitignore stats and a code-base outline.
- **Markdown generator** – `buildCompareMetricsMarkdown` (R146-R172) outputs a table of added/removed/modified counts and a sample list of changed paths.
- **Mermaid visualizers** – `changeKindPieMermaid` (R175-R185) and `depthBarMermaid` (R188-R204) emit Mermaid syntax for pie and bar charts of change kinds and path depth.

Impact

- **Performance** – All new functions process `rows` in a single pass, giving $O(n)$ time and $O(k)$ space where k is the number of distinct extensions, depths, or roots.
- **API surface** – The module exports only the new interfaces and functions; existing imports remain unchanged.
- **Observability** – Generated Markdown and Mermaid strings can be logged or displayed in UI components, aiding diff-report debugging.

Risks & follow-ups

- **Type alignment** – Verify that `ChangeKind` and `FileChange` in `../types` expose `kind` and `path` as expected.
- **Sample cap** – `SAMPLE_CAP` is hard-coded to 120; assess if this suffices for typical workloads.
- **Gitignore handling** – `metricsToPromptContext` includes a gitignore line only when `gitignore.skippedGitignore > 0`; ensure callers supply a non-null object when relevant.
- **Mermaid syntax** – The output uses Mermaid v9

- features (`xychart-beta`); test rendering in the target environment to avoid runtime errors.